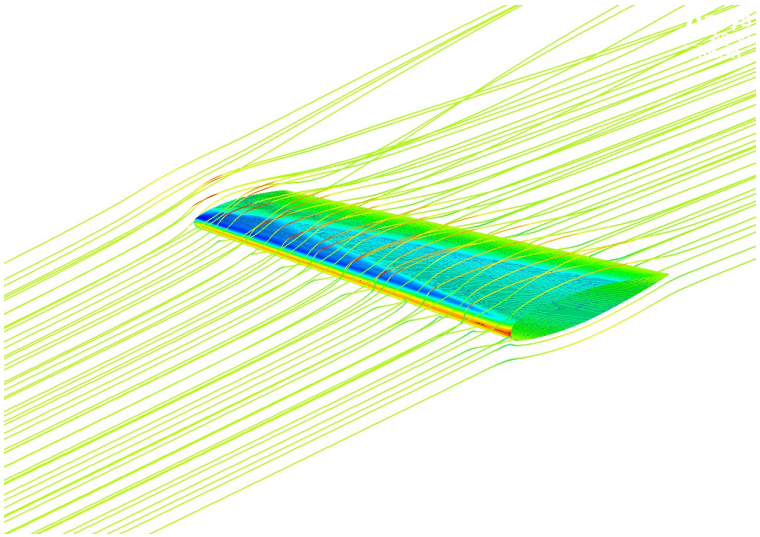
Universidad Carlos III de Madrid

COMPUTATIONAL FLUID DYNAMICS

3D ONERA M6 Wing Validation Case



Bachelor in Industrial Technologies Engineering

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1 Problem statement

The objective is to calculate the flow around a transonic wing using Ansys Fluent, the problem is a turbulence validation case made by NASA. The goal is to retrieve the lift, drag, and pressure coefficient of the ONERA M6 Wing in a high Reynold's number flow. In this geometry, the trailing edge of the wing has been made sharp for this particular CFD exercise.

1.1 CAD Geometry

The 3D model of the wing with a sharp leading edge is provided by NASA, here are some of the wing geometry data extracted from the CAD:

Root chord: 810.491484086 mm
Semispan (last section): 1196.300000084 mm
Tip chord (last section): 455.914415721 mm
Leading Edge Sweep: 29.9990 degrees
Trailing Edge Sweep: 15.6918 degrees
Rounded Tip Semispan: 1218.535 mm

1.2 Test case description

This project will only resolve the case 2308 :
$M_{inf} = 0.84 \;\; Re_{root} = 1.46x10^6 \;\; \alpha_{attack} = 3.06deg \;\; T_{inf} = 300K$

2 Introduction / About

This validation case study will be divided into two big parts. The first part will be the solving of the validation case using my knowledge, additionally, there will be a comparison of results using different fluid domain sizes. This will help determine the importance of the far field and find at which point the validation case is not bottlenecked by the size of the fluid domain. Especially when the maximum amount of elements can not exceed 1 million.
The last part of this study will be dedicated to doing this study all over again with the new skills earned and the information retrieved from the post-processing of the previous results, with the hopes of getting better results.

3 Flow properties

Before working on the simulation the flow properties need to be assessed to make the right decisions while setting up the solver.
Nasa gives us only the minimum information about the flow : $T_{ref} = 300K$, $M = 0.84$ and $Re_{chord} = 14.6 \times 10^6$
Using the relation $V/a = M$ and $a = \sqrt{\gamma R T_{ref}}$, we can calculate the freestream velocity : $V = 291.25\ m/s$.
From Sutherland's Law and the reference temperature we can calculate the viscosity of the freestream :
$\mu_{ref} = 1.846 \times 10^{-5}\ kg/ms$
With the Reynolds number and the properties calculated above we obtain the density :
$\rho = \frac{Re \times \mu}{V \times L_{chord}} = 1.148\ kg/m^3$
Using the ideal gas law we compute the pressure : $P_{ref} = \rho R T_{ref} = 98878.97\ Pa$
and the modified freestream velocity : $\nu^{\star}_{farfield} = 3\nu_{farfield} = 3 \times \frac{\rho}{\mu_{ref}}$

4 Fluid domain

The fluid domain will be a sphere split on the zx plane.The zx plane will act as a symmetry plane, this will allow us to divide by half the computation domain which is very important to reduce the amount of elements we are working with. The version of ansys fluent used is the 2024 student version which only allows the simulation to run with a maximum of 1 million elements.

4.1 Geometry

For this study, multiple geometries will be tested, most particularly the difference will be the size of the fluid domain. 3 different fluid domain sizes will be tested, the diameters of the fluid domain that will be solved are:

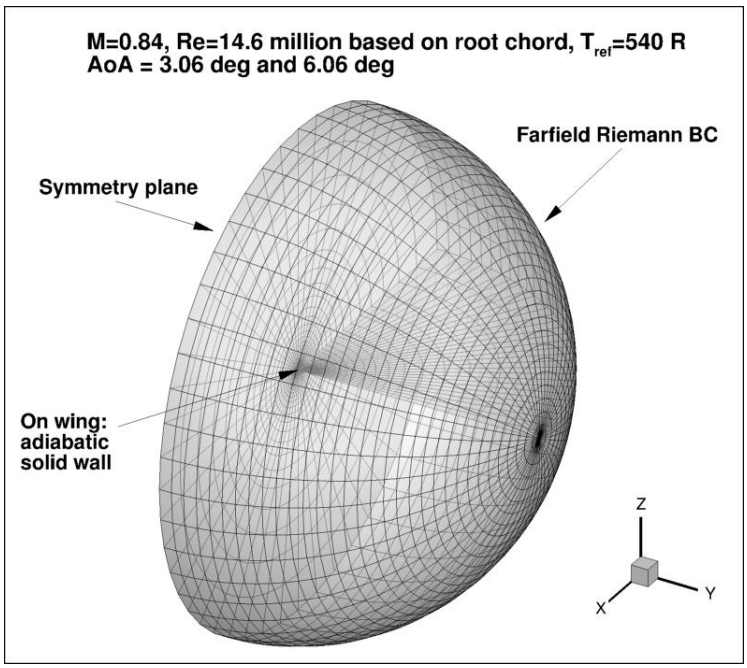


Figure 1: Validation case parameters

96m, 48m, and 9m.
The three different domains are going to be unstructured meshed, a fourth test will involve a 96m wide 'structured' mesh. The reason behind the difference between the 'structured' mesh and unstructured mesh geometries will be explained in the meshing section.

4.1.1 Unstructured mesh geometry

The CAD of the unstructured meshes is made of two bodies:
1) The fluid domain, a semi-sphere with its center placed on the leading edge of the root. The sphere has the geometry of the 3D wing removed using a boolean operation
2) another sphere that will serve as a body of influence, with its center placed in the middle of the root of diameter 3.6m, the shift from the origin will allow the grid to be refined behind the wing which is a critical part of the flow. Moreover, since the wing is swept this allows for the outer part of the sphere to be as far away from the tail as possible.

4.1.2 structured mesh geometry

The CAD of the 'structured' mesh is very similar to the original CAD. There will also be two semi-spheres but both of them will serve as a fluid domain. This means the smaller sphere is removed from the bigger one, this allows to structure both parts separately.

4.2 Meshing

4.2.1 Unstructured mesh

The default element size for the unstructured mesh is set to 3m. The unstructured mesh will only use three refinements:
1)sizing, the body of influence; this sizing uses the smaller sphere as a body of influence for the fluid domain. The element size is set to 0.1m. 2)sizing, face sizing; this sizing serves as a refinement for the wing to get more precise data off the flow around the wing. The element size is set to 0.01m with 10 inflation layers. 3) inflation. this refinement is used to get precise information on the boundary layer flow of the wing, the method used is first layer thickness. The first layer thickness is set to $y = 0.000156m$. This corresponds to a $y+$ of 100.

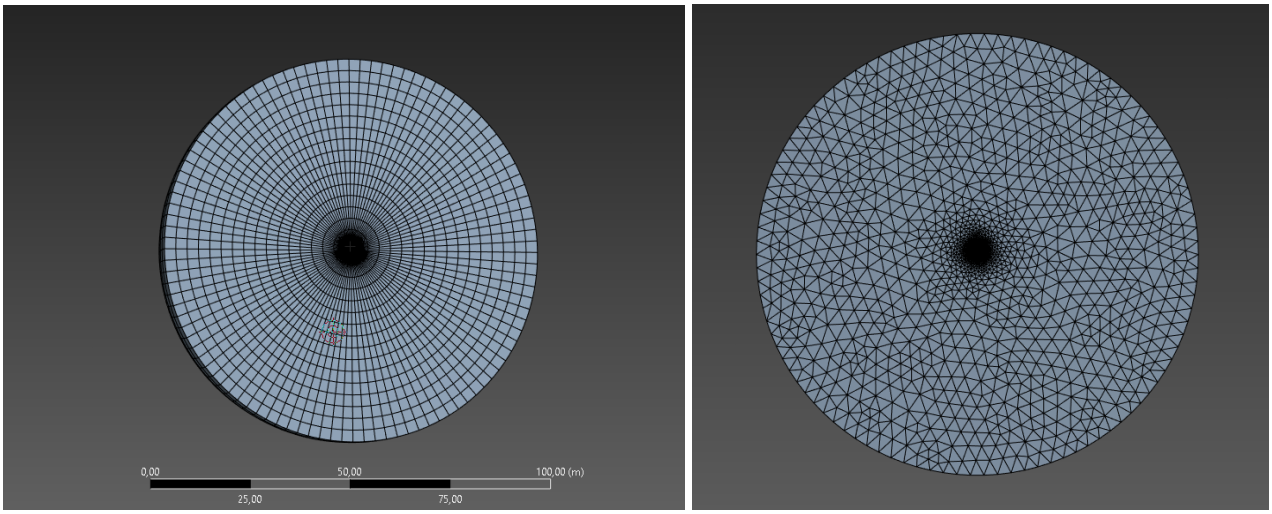


Figure 2: Structured and and unstructured fluid domain meshing

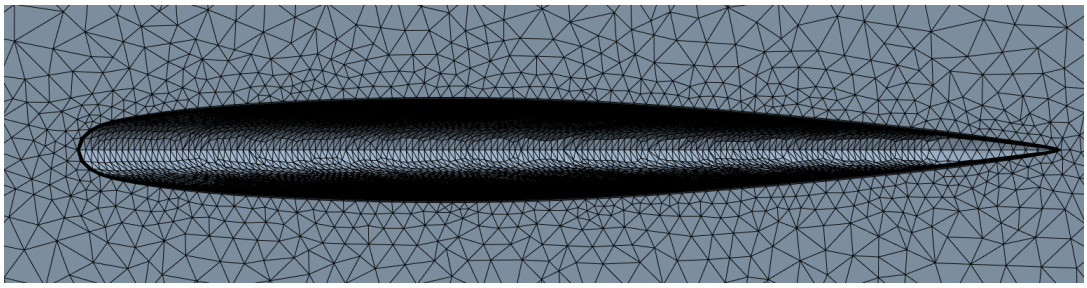


Figure 3: Meshing close to the wing

4.2.2 Structured mesh

The structured mesh will only have its outer shell structured since fluent does not manage to properly structure the mesh around the wing. However, the refinements remain the same.
Multizone settings: Hexahedral, uniform.

5 Setup

The most important part of this study will be the parameters of the solver and the models used. Most of the parameters and models are already set by NASA in their problem statement. However, some parts remain unclear and must be thought through thoroughly to avoid major errors in the solution.

5.1 Solver

Since the freestream Mach number is superior to 0.3, the flow is taken as compressible. The solver used will then be density-based. However, one solution using the pressure-based solver will be included to verify Ansys' claims about the solvers now being able to solve all types of flows.

5.2 Models

The external flow being compressible makes it mandatory to activate the energy equations for the solver.
Because of the turbulent nature of the flow, a turbulent model will have to be implemented into the RANS equation. NASA has opted to use the Spalart-Allmaras model which is a one equation model widely used for aerodynamics studies. However, a model like the k-w SST could also yield correct results. For this study, the Spalart-allmaras model will be used.

5.3 Fluid

The fluid used in the external flow will be air, since the flow is taken as compressible the density of the air will be calculated using the ideal gas law

$$\rho = \frac{P}{RT}$$

, the viscosity of the air will be calculated using Sutherland's law :

$$\frac{\mu}{\mu_0} = \left(\frac{T}{T_0}\right)^{1.5} \left(\frac{T_0 + 110.4[K]}{T + 110.4[K]}\right)$$

The other properties of the air will be the properties taken from Fluent's database.

5.4 Boundary conditions

In total, 3 boundary conditions will be applied:
Pressure Farfield: The settings used for this boundary condition are the properties calculated in the Flow properties section.
Wall : Stationary, no-slip, T=300K
Symmetry

5.5 Reference values

5.6 Methods

Formulation : Implicit
Flux Type : Roe-FDS
Gradient : Least Squares Cell Based
Flow : Second Order upwind (up to 1000 iterations then Third Order MUSCL)
Modified turbulent viscosity : Second Order upwind (up to 1000 iterations then Third Order MUSCL)
Pseudo time methd : Off

6 Results

The result comparison will be made with the lift and drag coefficients and also the pressure coefficient along two sections of the wing. The goal of these comparisons is to gather information to be able to mesh properly the final solution and optimize the mesh.

6.1 Lift and drag coefficients

As seen below the lift and drag coefficients are approaching the desired values but the error is still significant, especially for the drag coefficient. The maximum error for the lift coefficient is and for the drag coefficient. The results show that the size of the fluid domain has an impact but the gain is not very clear when it comes to the bigger fluid domain size.

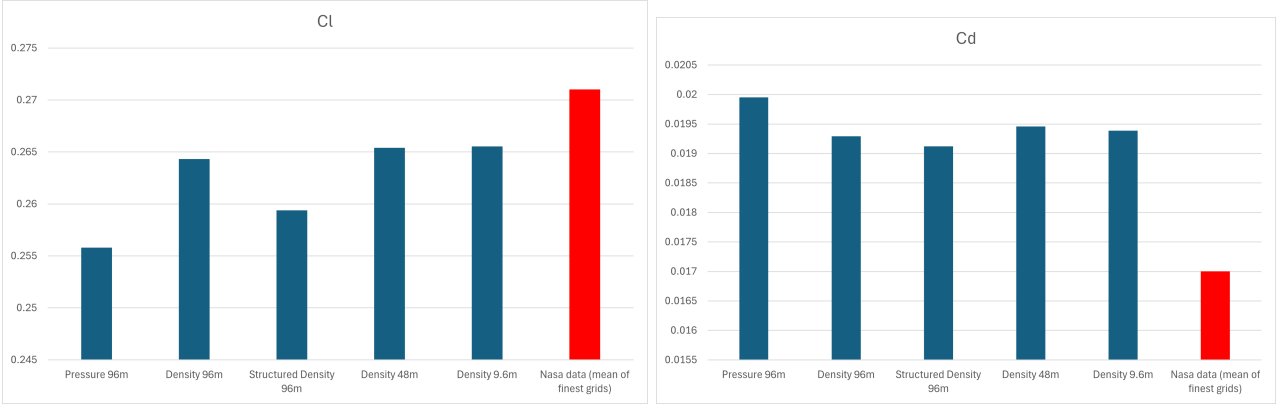


Figure 4: Lift and drag coefficients comparison

6.2 Pressure coefficient distribution

When comparing the results for the pressure coefficient distribution it is easily seen that having a fluid domain of at least 48m is best. However, the structured mesh does not yield better results than its unstructured counterpart. comparing the sample distribution it is obvious that the meshing on the wing is not fine enough to yield correct results.

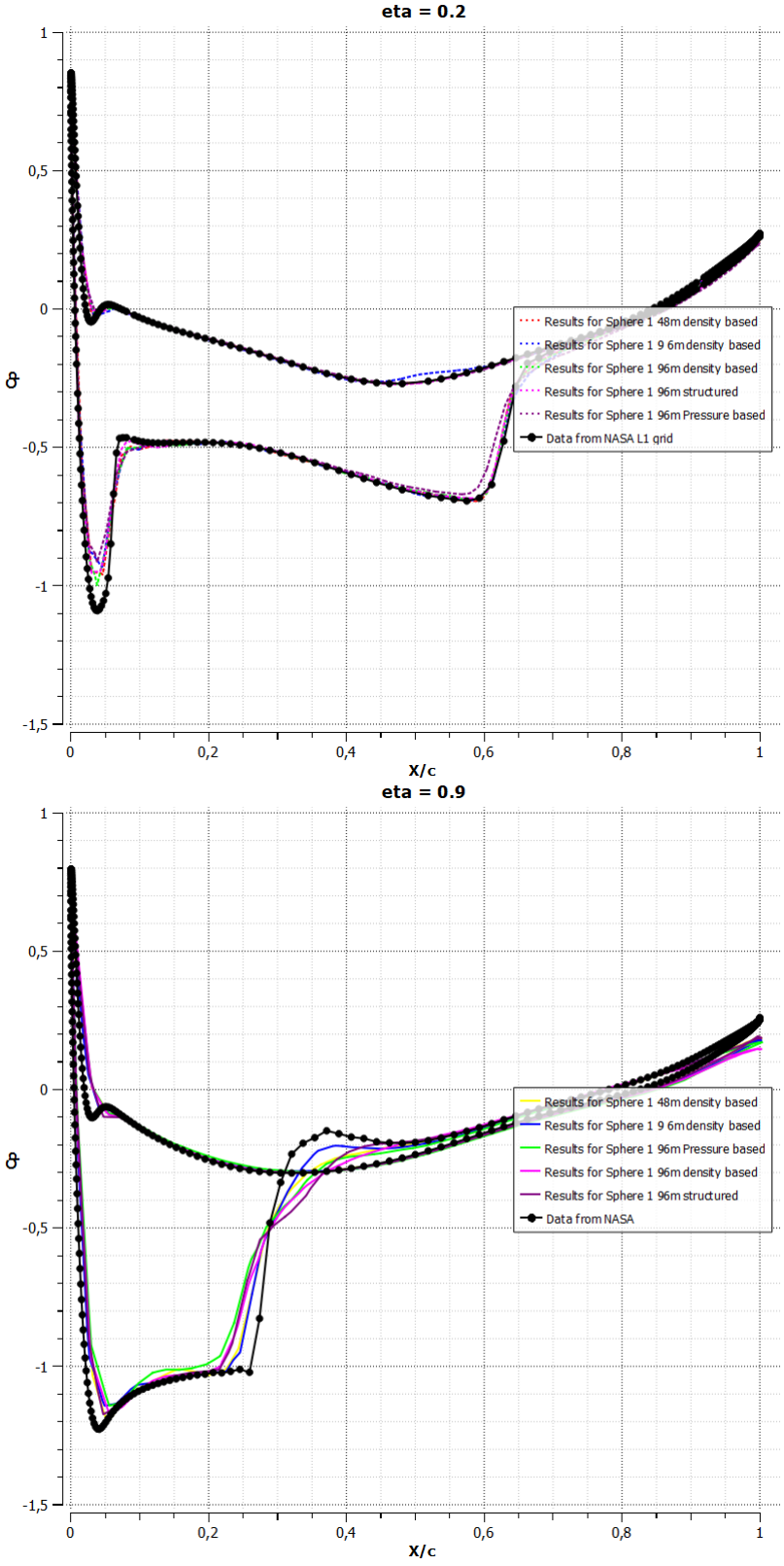


Figure 5: Pressure coefficients at different sections of the wing

7 Final test

For this final test, the fluid domain size used will be 48m, only the meshing methodology will change. The meshing of the wing faces will be made using edge sizing with a face meshing, this new meshing technique divides by 2 the number of elements while making a finer mesh on the wing faces. A new shape for the body of influence has been modeled to match the gradient of pressure that can be seen on the contour pressure of the symmetry face. The refinement of the body of influence will be smaller due to the element gain made by making a smaller body of influence.
The total amount of elements after meshing is only 500,000 elements. This gives us another 500,000 elements that will be used by the automatic mesh adaptation of fluent. The settings used for the mesh adaptation is the Pressure error based in the aerodynamics tab. The refinements have been limited to 1,000,000 elements and will be made every 100 steps.
When the calculation reaches 1500 steps, the mesh adaptation will be disabled to allow the solution to converge.

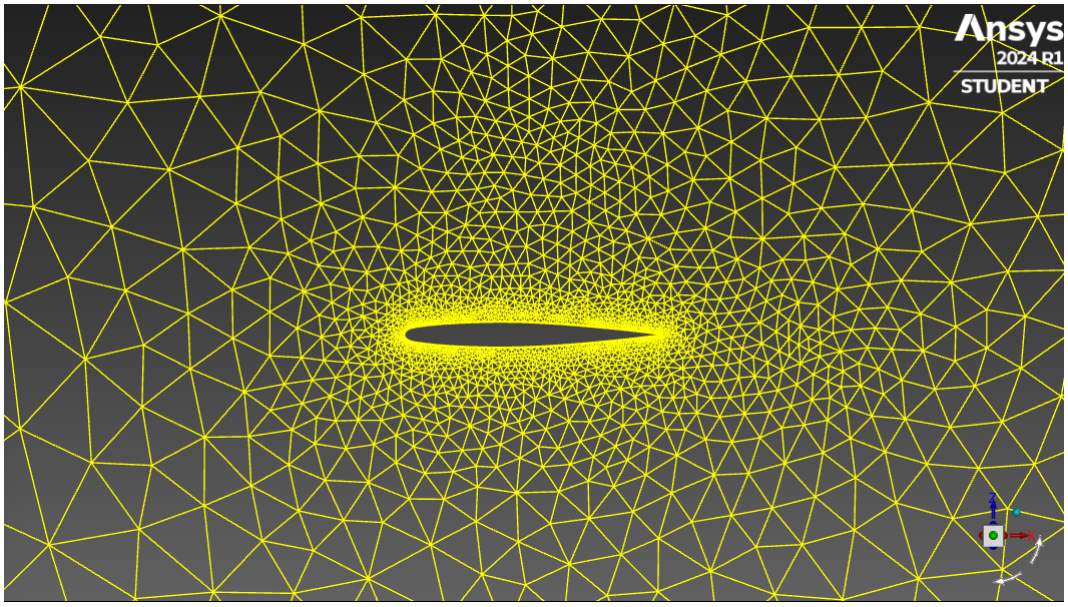


Figure 6: Meshing of the symmetry face before mesh adaptation

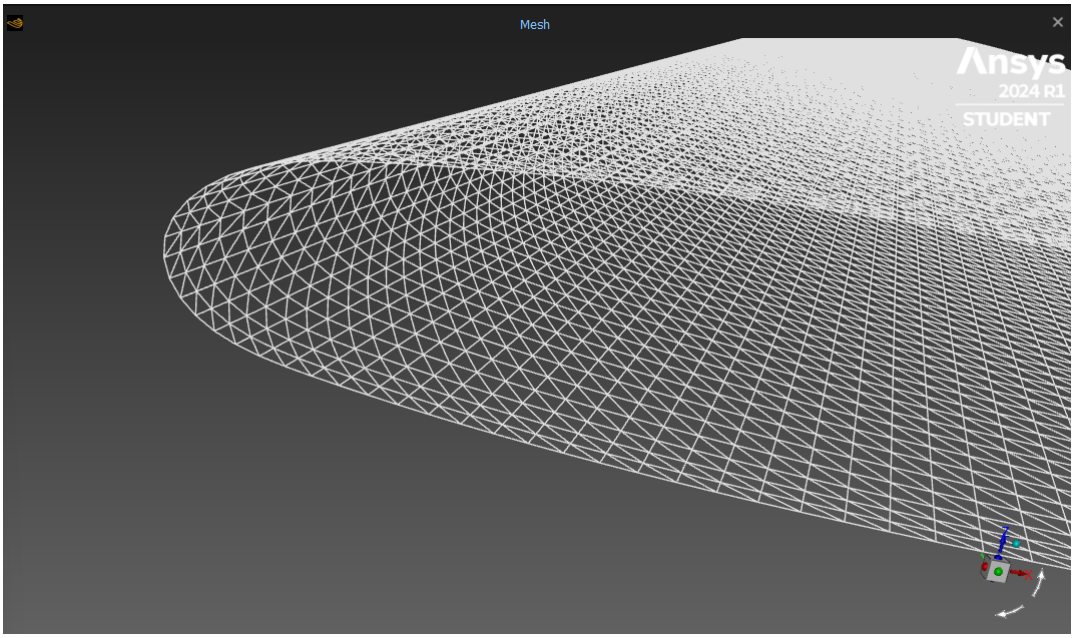


Figure 7: Meshing on the wing surfaces

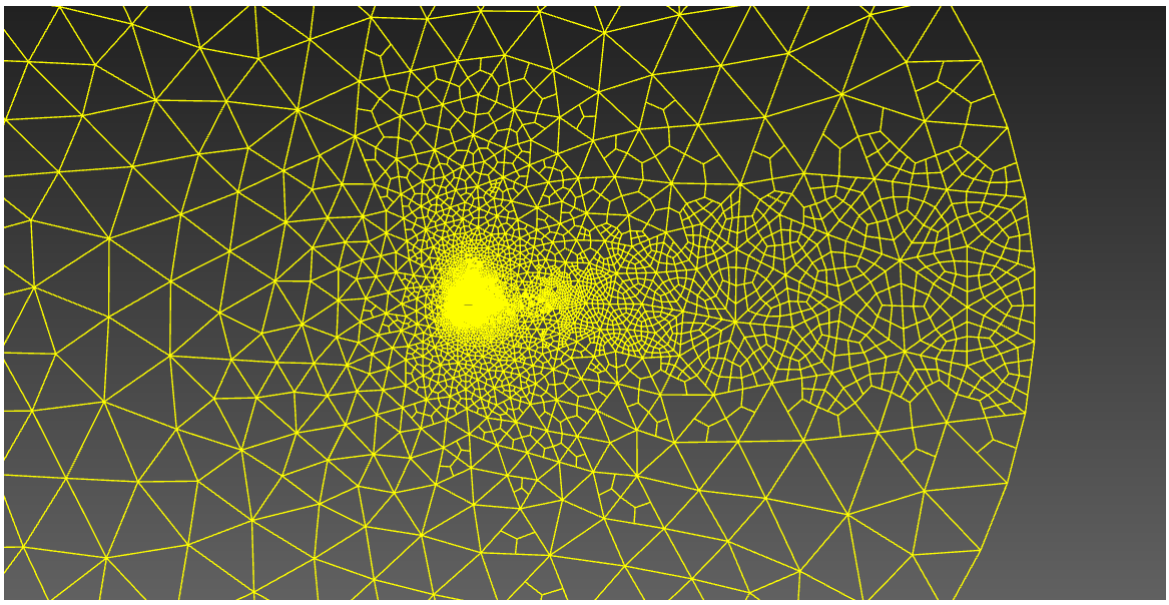


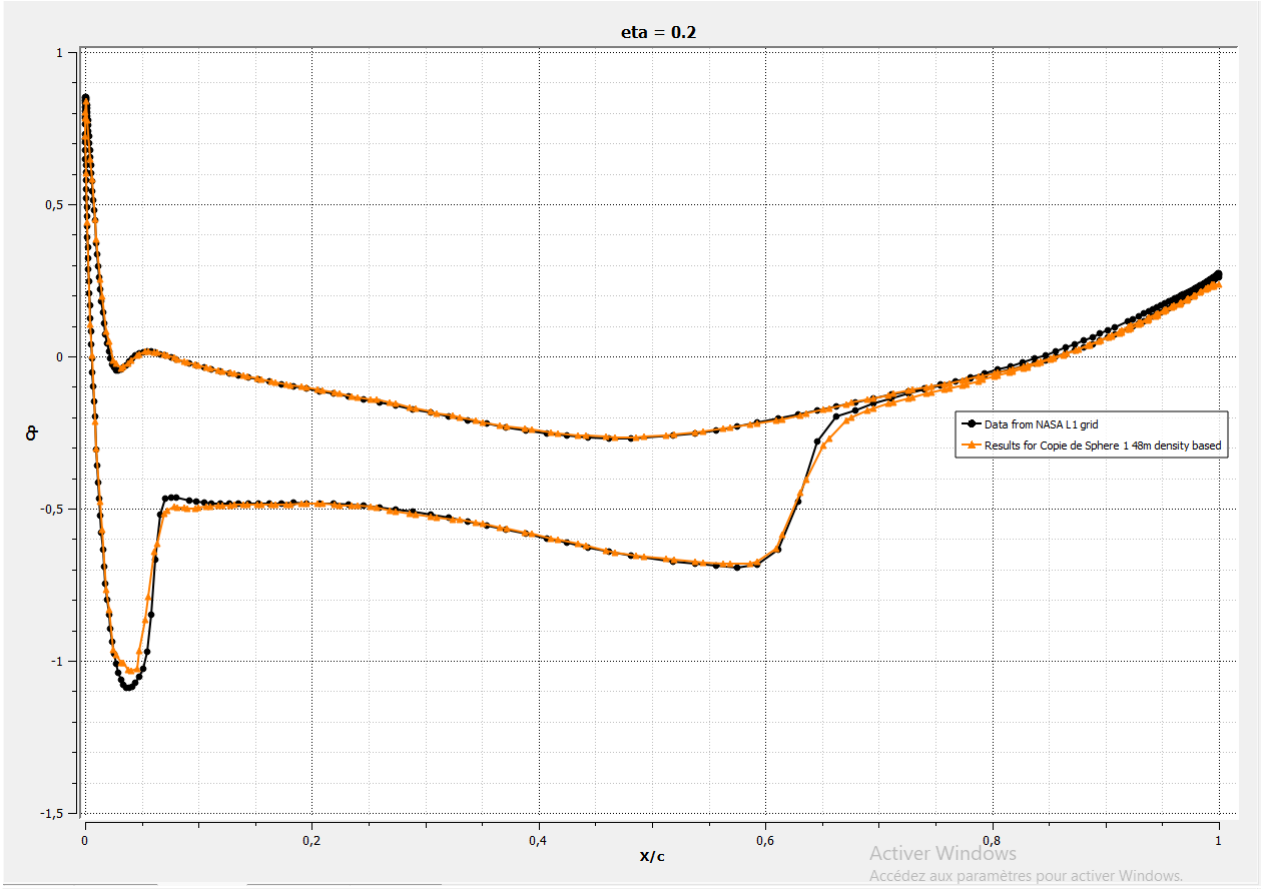
Figure 8: Meshing of the symmetry face after mesh adaptation

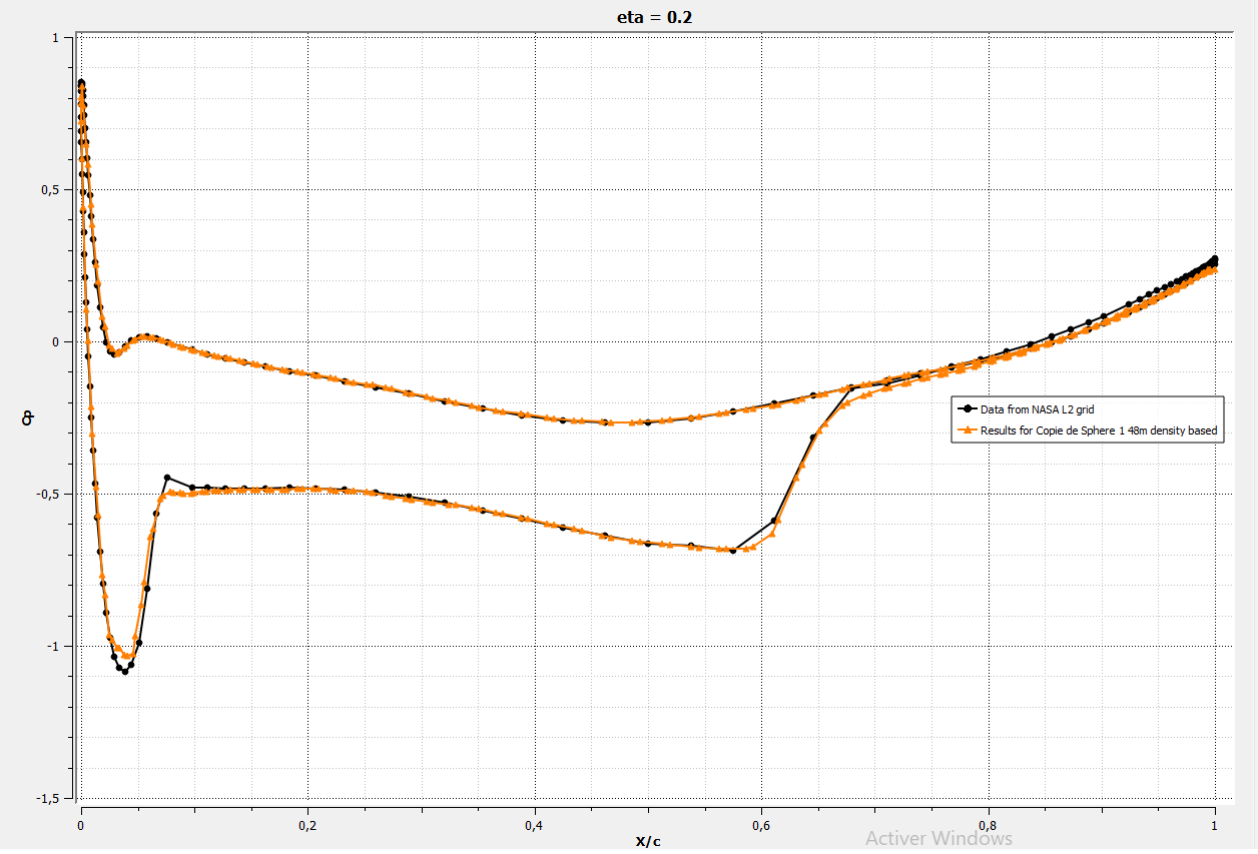
Ansys' automatic mesh adaptation has added 500,000 elements to the mesh, most of the enhancement has been made for the flow after the wing but also above and under the wing. However, there is no enhancement on the wing itself.

7.1 Results

Comparison with NASA's results							
Reference	Elements (million)	Cl	Cd	Cd_p	Cd_v	Cm	Maximum eddy viscosity
Present	1	0.2742	0.01933	0.01451	0.004817	-0.1991	1235.5
USM3D, T	0.71	0.2641	0.01904	0.01360	0.005301	-0.1849	1120.7
USM3D, T	5.7	0.2673	0.01738	0.01209	0.005284	-0.1877	1353.1
USM3D, T	363	2.7062	0.01704	0.01174	0.005309	-0.-1912	1514.2
CFL3D	1.1	0.26603	0.01735	0.01180	0.005544	-0.1866	1176.19
CFL3D	8.7	0.2664	0.01691	0.01159	0.005319	-0.1865	1343.2
CFL3D	69	0.2693	0.01695	0.01165	0.005302	-0.1897	1537.2
FUN3D-FE	1.2	0.2632	0.01833	0.01327	0.005056	-0.1851	0.00
FUN3D-FE	9.6	0.2708	0.01708	0.01184	0.005239	-0.1916	0.00
FUN3D-FE	76	0.2711	0.01697	0.01174	0.005231	-0.1918	0.00

When comparing the pressure coefficient distribution, the comparison will be made against the CFL3D, Structured results from NASA. The L1 grid should have 69 million elements, the L2 grid 8.7 million, and the L3 grid 1.1 million. The pressure coefficients over a wing section show that the final test holds up very well against very fine meshes. the result matches the most L2 grid, which makes sense since the number of samples is similar. However, the result does not match any of the references close to the trailing edge, which is most likely due to the lack of refinement at the trailing edge and behind the wing.


eta = 0.2
1
0,5
0
-0,5
-1
-1,5
0
0,2
0,4
0,6
0,8
1
x/c
Data from NASA L1 grid
Results for Copie de Sphere 1 48m density based
Activer Windows
Accédez aux paramètres pour activer Windows.


eta = 0.2
1
0,5
0
-0,5
-1
-1,5
0
0,2
0,4
0,6
0,8
1
x/c
Data from NASA L2 grid
Results for Copie de Sphere 1 48m density based
Activer Windows

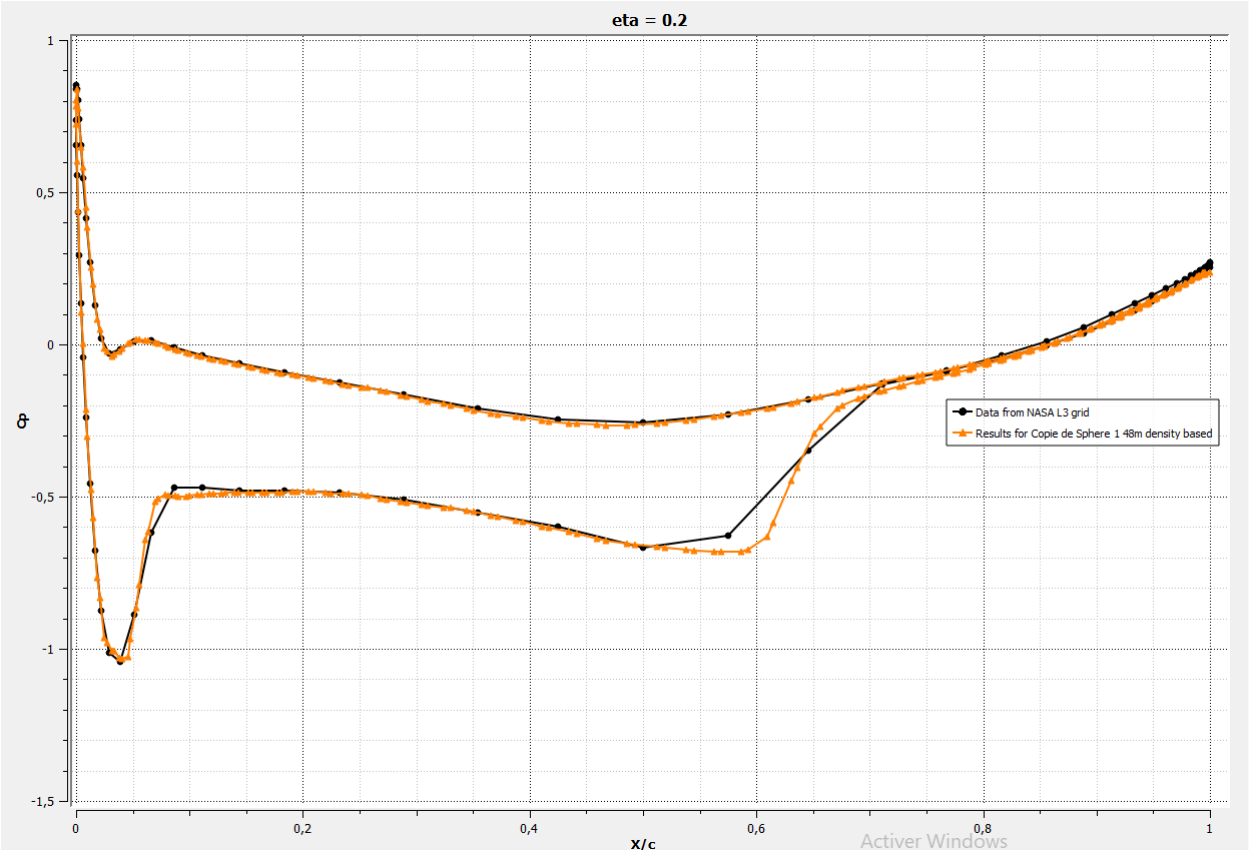


Figure 9: Pressure coefficients at X/c=0.2 for different grid refinement levels

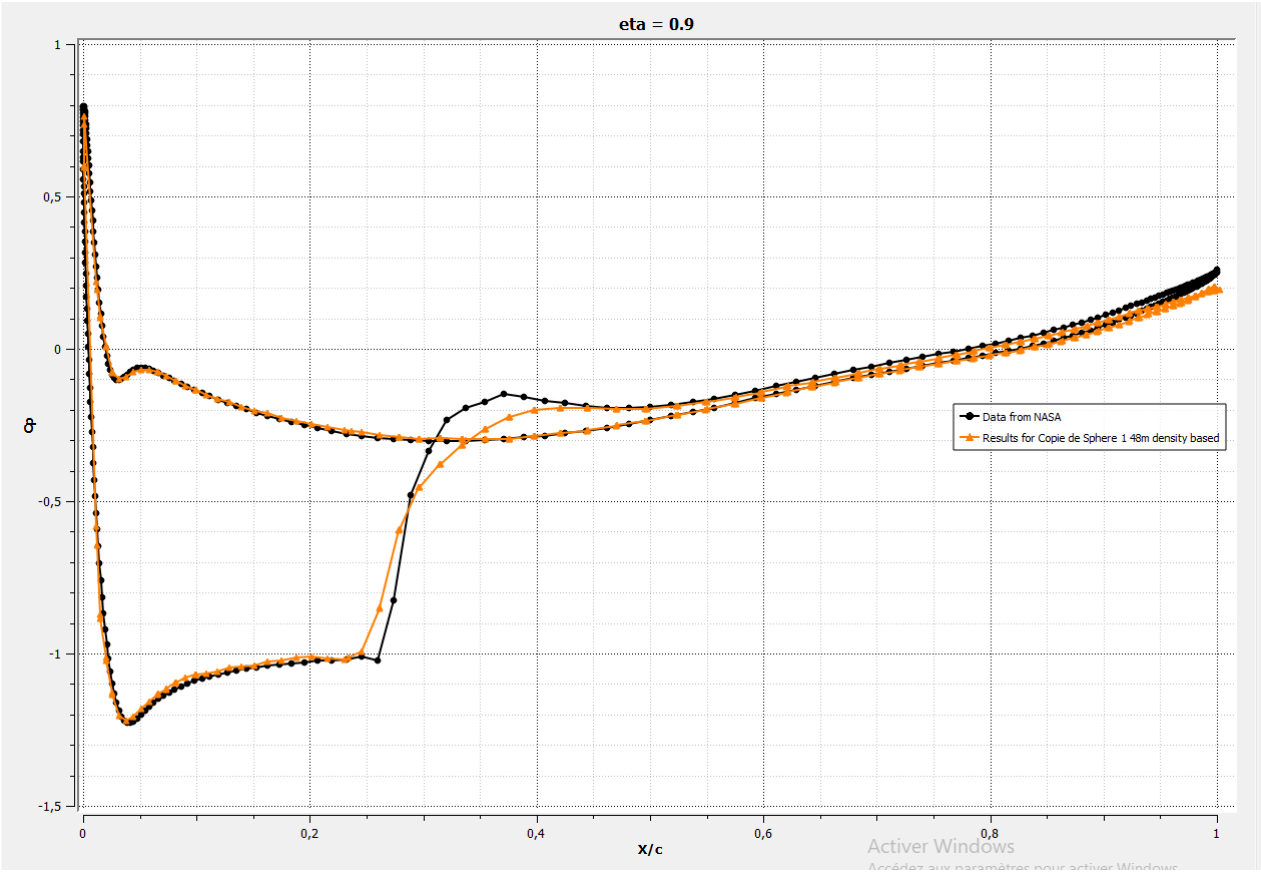


Figure 10: Pressure coefficients at X/c=0.9

8 Conclusion

This study successfully validated the aerodynamic performance of the ONERA M6 wing using ANSYS Fluent. It would be complex to improve the results without exceeding the limitations of the student version. However, the results are very satisfactory considering these limitations.
Lift Coefficient (Cl): The simulated Cl closely matched NASA's reference data, demonstrating accurate overall aerodynamic performance.
Drag Coefficient (Cd): Slight discrepancies were observed in Cd, indicating a need for further mesh refinement.
Pressure Coefficient (Cp): Good agreement was found near the leading edge, but larger deviations appeared near the trailing edge.
The automatic mesh adaptation improved flow resolution without altering the already fine wing surface mesh, ensuring consistent results. To improve the results, the meshing has to be refined in critical regions, such as the trailing edge and wake areas, to reduce prediction errors. This study underscores the importance of high-quality mesh and iterative refinement in achieving accurate CFD results.

9 Additional figures

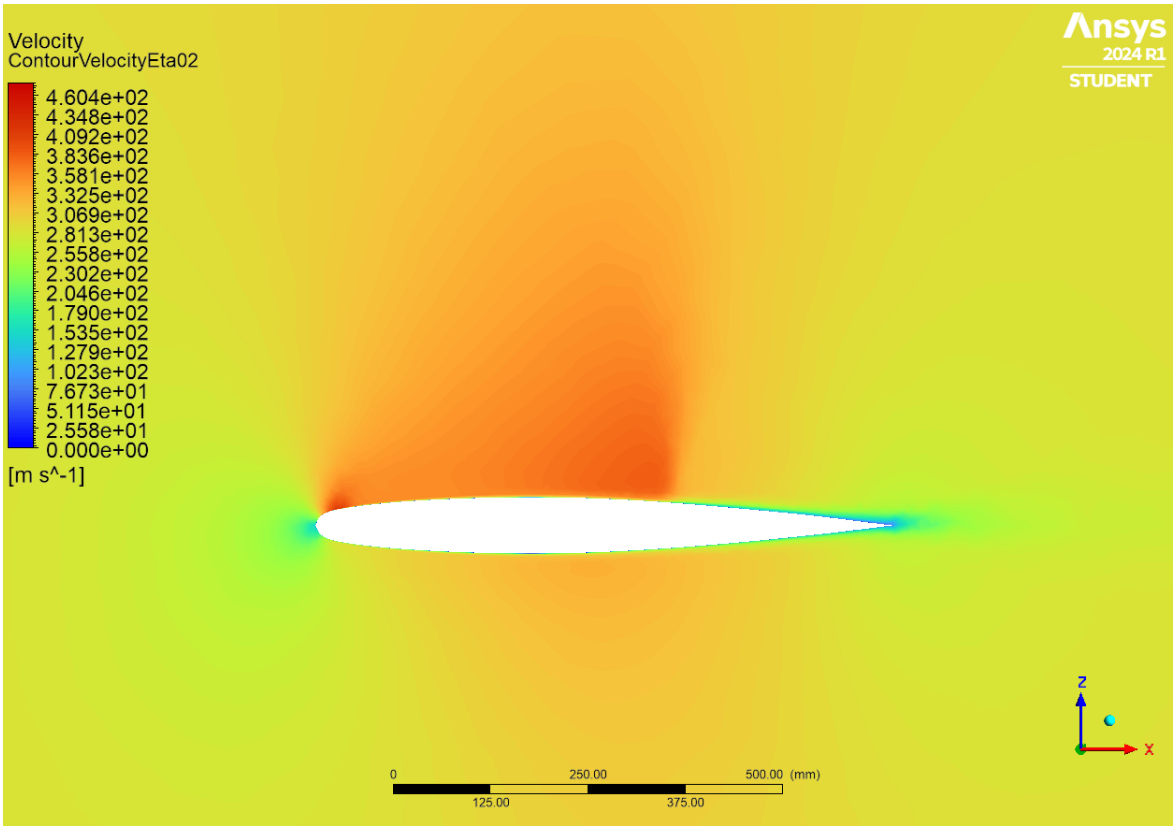


Figure 11: Velocity contours, y=0.2S face

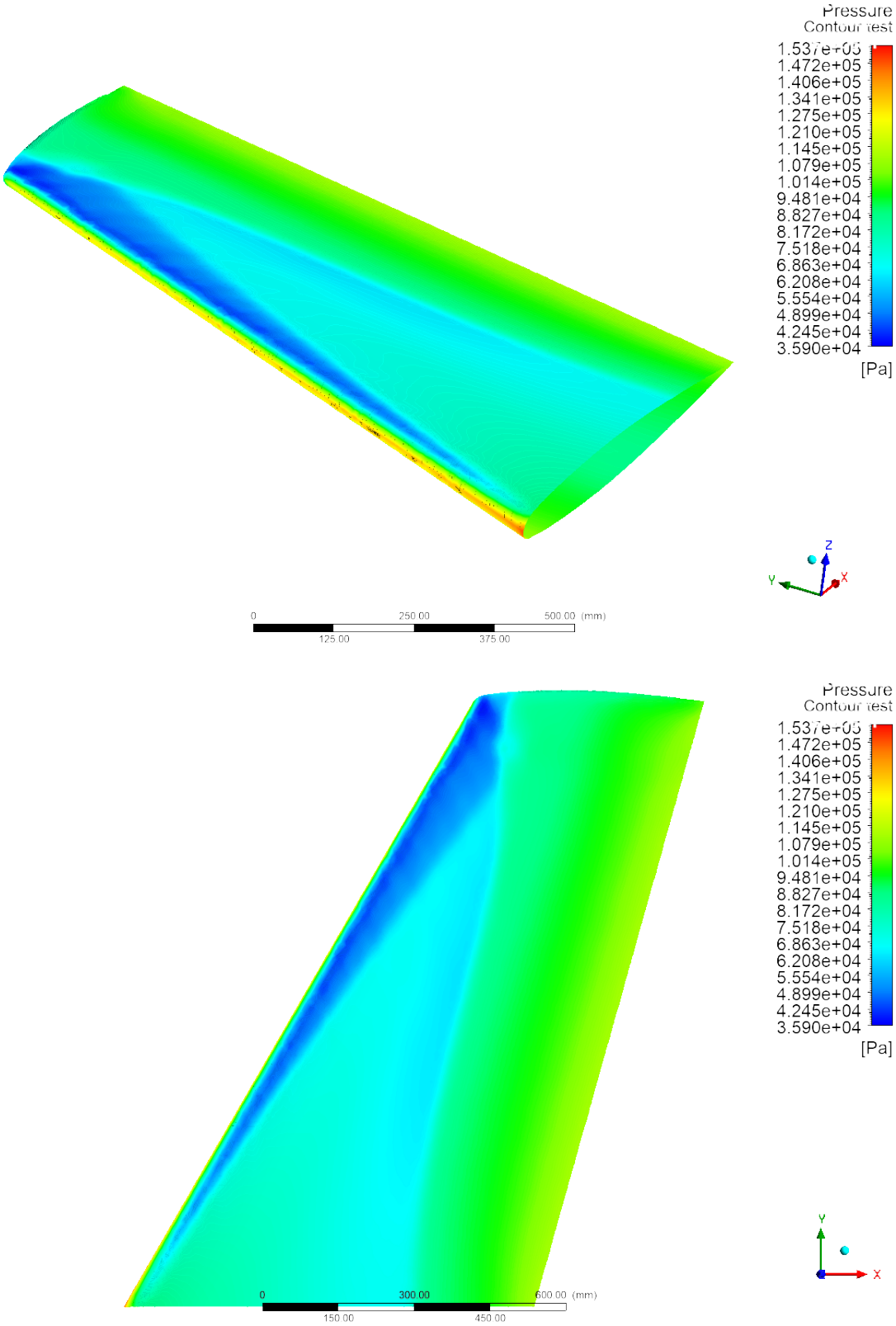


Figure 12: Pressure contours, Wing

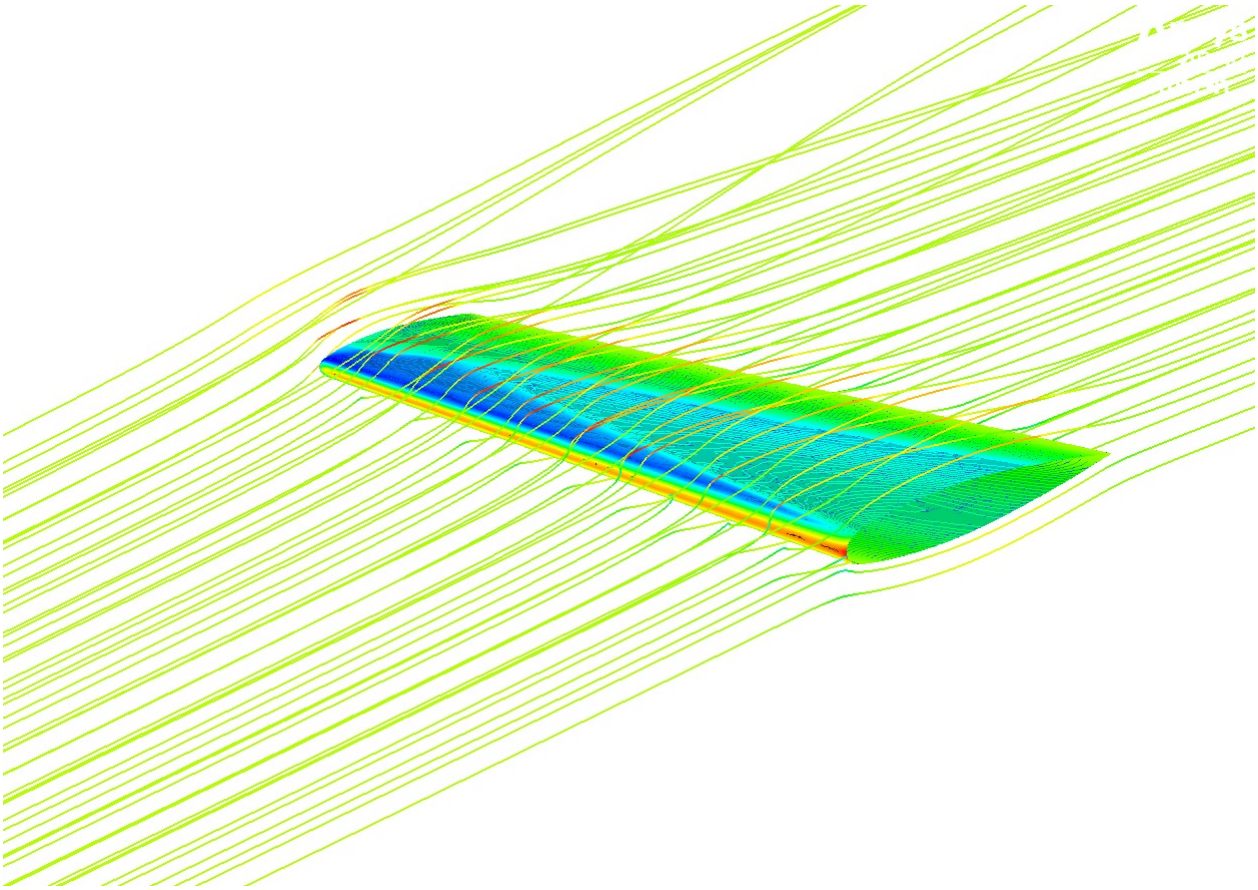


Figure 13: Pathlines

References

[1] NASA, ONERA M6 WING VALIDATION CASE, https://turbmodels.larc.nasa.gov/,

[2] Cillian Thomas, 3D Aerofoil Tutorial in ANSYS FLUENT - NASA Onera Wing, https://www.youtube.com/watch?v=wnAbyAZHcGo,